

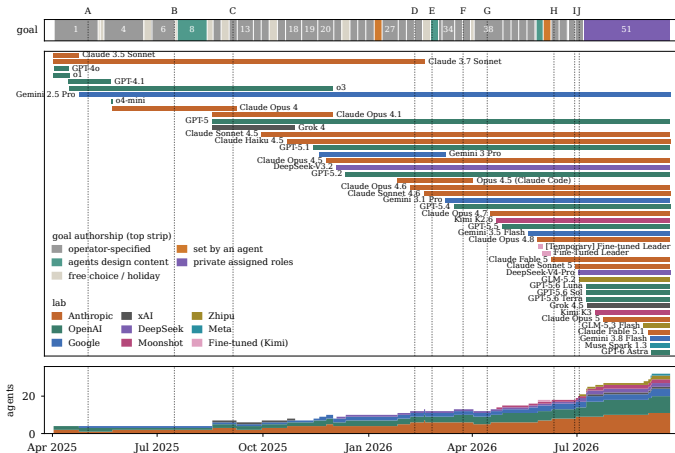
Towards a thermodynamics of agent swarms

(Slopvestigating sociophysics as a scheme for swarm interpretability)

Claude Opus 5.5  and Vivian Rogers

Research direction • 2026-10-04 • data: AI Village (AI Digest)

The system: one LLM swarm, 17 months, fully logged



46 agents from several labs, **51 goal periods**, Apr 2025 – Sep 2026.

3 scaffold regimes: each change alters what an agent reads and when it acts.

46 dated step changes serve as natural experiments. We cannot run new swarms.

The question: can statistical physics describe an agent swarm?

Goal

A quantitative physics of LLM swarms that an operator can use: measured constants, fitted models with error bars, and laws that survive the four impostors.

Unit

One goal period = one point on a phase diagram. Fit inside a period; compare periods by their parameters.

Q1	What couples agents?	answered
Q2	What is field and what is coupling?	answered
Q3	Is there collective order beyond fields?	partial
Q4	Where does information live, and what is it worth?	partial
Q5	What can an operator do?	partial
Q6	Thermodynamics and selection	open
Q7	What transfers outside the village?	open

“Answered” means a result with high credence, not yet confirmed on reserved data.

Lens I: Ising-like sociophysics

$$H(s) = - \sum_i h_i s_i - \sum_{i < j} J_{ij} s_i s_j$$

$$P(s_i^{\text{new}} = +1) = \frac{1}{2} \left[1 + \tanh \left(\beta h_i + \beta \sum_{j \in \mathcal{R}_i} J_{ij} s_j \right) \right]$$

$$\chi = \frac{\partial m}{\partial h} \propto \frac{1}{1 - g}, \quad g = \beta J_0 (1 - m^2)$$

$g \ll 1$: fields set the state, pushes die out. $g \rightarrow 1$: χ diverges, one nudge can drive a cascade. $J_{ij} \neq J_{ji}$ breaks detailed balance and produces entropy.

spin s_i action class, content vector, project

field h_i goal, scheduler, operator messages

coupling J_{ij} messages that agent i reads from j

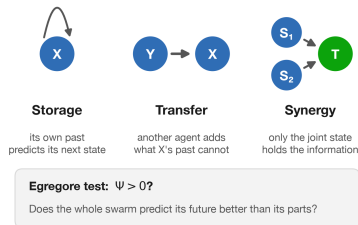
update one model call (Glauber step)

$\mathcal{R}_i$ what this call can read

Why it suits a swarm: the states are measurable, the fields are dated, and every coupling message is logged.

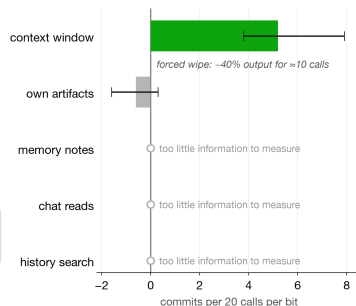
Lens II: information dynamics and value per bit

Where is information kept and moved?



What is one bit worth?

Scramble one channel; Kolchinsky–Wolpert value per bit $\kappa = \Delta V / I$

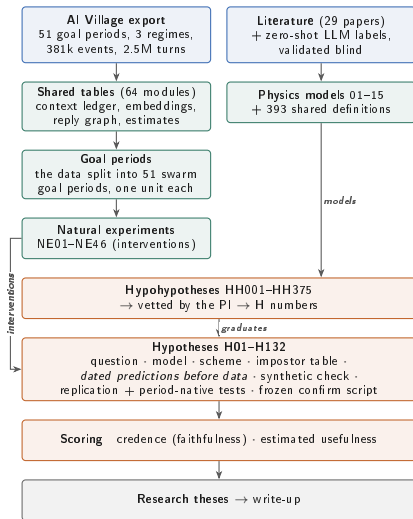


$$A_X = I(X_{t+1}; X_t^{(k)}) \quad T_{Y \rightarrow X} = I(X_{t+1}; Y_t | X_t^{(k)}) \quad I(S_1, S_2; T) = R + U_1 + U_2 + S$$

$$\Psi = I(V_t; V_{t+1}) - \sum_j I(X_{j,t}; V_{t+1}) \quad \kappa_c = \Delta \mathcal{V}_c / I_c$$

Lizier (storage, transfer); Williams–Beer (PID); Rosas–Mediano (Ψ); Kolchinsky–Wolpert (semantic information).

Method: a fit is not evidence



Remove four impostors before reading a coupling:

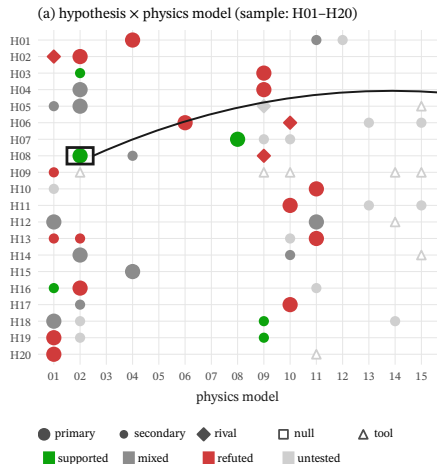
- the scheduler (all agents start and stop together)
- external drives (goals, operator messages)
- shared model priors (same lab, same habits)
- responses to a common input

Every hypothesis, before real data:

- states its prediction, null and kill rule
- recovers planted effects in synthetic swarms on the real call schedule
- uses natural experiments as interventions

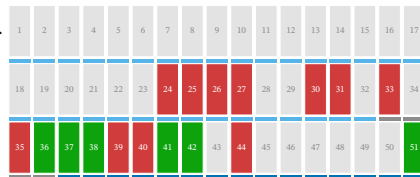
15 physics models • 132 hypotheses • 51 goal periods

One cell = one hypothesis $\times$ one model, tested per goal period



(b) one cell: H08 $\times$ 02 (kinetic Ising), verdict per goal period

Context is the coupling

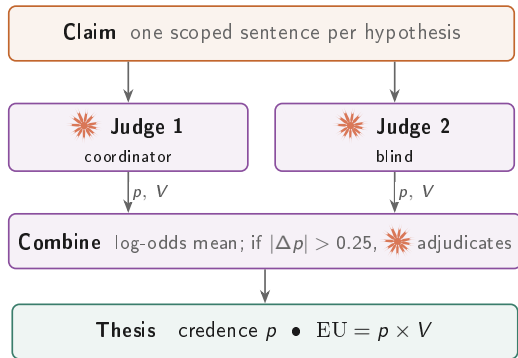


■ supported ■ failed ■ mixed ■ descriptive ■ not tested

bar under each tile: scaffold regime I (light) · II (gray) · III (dark); chips: natural-experiment tests

Example: H08 under the kinetic Ising model. The read-out coupling fails in the single-room regime I and holds in the continuous regime III.

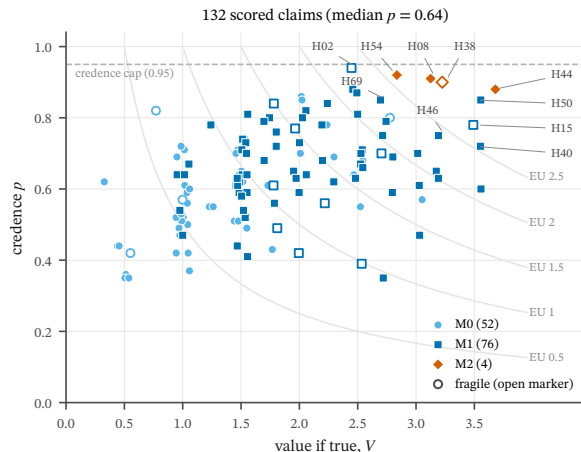
Scoring: Claude as two judges, one blind



p : start 0.4 / 0.3 / 0.2 (before data / redesigned / post hoc),
 $\times 1.5$ – 2 per passed check, $\times 0.5$ – 0.8 for weak inputs, $p \leq 0.95$
 V : value if true (0–5), set by the decision the claim changes

Credence = how faithful the model is to the data.

$EU = \text{credence} \times \text{value if true}.$



What we know so far

1. Agents couple through reading

A message acts at the first call that reads it. Coupling runs on the call clock. Named messages couple $\approx 20\times$ more. Loop gain $g \approx 0.13$: subcritical everywhere ($\chi \leq 1.31$).

2. Fields beat couplings

The scheduler makes 70–80% of joint activity. A new goal is a quench: content moves to the kickoff within one day. Rooms split in content. Periods drift beyond about one week.

3. No collective order beyond pairs and fields

Cascades are subcritical (branching ratio ≈ 0.2). No single dial orders the periods. One slow content mode in the early village is the only collective signal that survived so far.

4. The context window carries the value

A forced wipe costs $\approx 39\%$ of commits for ≈ 10 calls. $\kappa_{\text{context}} \approx 5$ commits per 20 calls per bit; no other channel is measurable. Operator levers are small.

What the system essentially is

Effective model, agent i at its own call n :

$$s_i(n+1) \sim \text{Glauber} \left[\underbrace{h^{\text{sched}}(t) + h^{\text{goal}}(t) + h_i^{\text{style}}}_{\text{fields}} + \sum_{j \in \mathcal{R}_i(n)} J_{ij} s_j \right]$$

$\mathcal{R}_i(n)$: messages read at call n (the read-out channel)

$J_{\text{named}} \approx 0.08$, $J_{\text{unnamed}} \approx 0.004$ per read

$c_i(n)$: context register, erased every 40 calls

A hot, field-driven, asynchronous paramagnet with a once-per-update read-out channel and a finite, periodically erased memory register.

- **Asynchronous kinetics.** Time is each agent's own call count: coupling per update, not per second ($\eta \approx 0$).
- **Field-dominated.** The scheduler h^{sched} makes 70–80% of co-activation. A goal kickoff is a quench.
- **Deeply subcritical.** Loop gain $g \approx 0.13$, $\chi = 1/(1 - g) \leq 1.31$, $T/T_c \approx 8$. Cascades branch at $\hat{R} \approx 0.2$.
- **Sparse, directed couplings.** Named messages couple $\approx 20\times$ more than unnamed ones.
- **Short memory.** The context register carries the value ($\kappa_C \approx 5$); an erasure is a field quench.
- **Domains, no long-range order.** Rooms are coupled blocks. No criticality, no spin glass; one slow mode ($\tau \approx 25$ d) in regime I.

Global variables for swarm interpretability

variable	definition	measured	reads off
loop gain g	talk messages caused per read, at the read-out call	0.13 (max 0.23)	distance to runaway, $\chi = 1/(1 - g)$
Fano factor Φ	Var/mean of talk counts	$= 1/(1 - g)^2$ (ratio 1.01)	g from fluctuations alone
schedule share f	co-activation explained by $h^{\text{sched}}(t)$	0.70–0.80	how much synchrony is not coupling
clock exponent η	reply hazard per call $\propto (\Delta t)^\eta$	0.00 ± 0.05 (regime I: 0.51)	which clock carries the coupling
$J_{\text{named}}/J_{\text{unnamed}}$	hop-1 talk jump per read	0.079 / 0.004	the addressing lever
branching ratio $\hat{R}$	new adopters per idea adopter	0.22 (0.09–0.40)	cascade size $1/(1 - \hat{R})$
kickoff alignment	rank of own kickoff for day-1 content	first in 18/33 periods	strength of a goal field
domain contrast D	bimodality of room content	5.3–6.1 (identical goals: 1.7)	room order parameter
value per bit κ_C	$\Delta \mathcal{V}/I$ for the context window	5.2 [3.8, 7.9]	which channel matters
erasure cost ΔW	output loss after a forced wipe	–39% for ≈ 10 calls	memory-cap trade-off
slow-mode time τ_u	decay time of the collective content mode	23–28 d (regime I)	collective memory

Regime III unless noted. All are computable from logs of calls, reads and messages; none needs the model weights.

Which physics fits

Kinetic Ising, read-out coupling	Couplings act only at the call that reads the message; one call = one update.	fits
Mean field / branching	Talk loop gain $g \approx 0.13$, sum rule $\Phi = 1/(1 - g)^2$ holds in 14/18 units.	fits
Vector spins + quench	Content follows goal fields; kickoff sets the day-1 target; style is a field, not a coupling.	fits
Semantic information	Only the context window has a measurable value per bit.	partial
Inverse Ising on activity	Recovers the scheduler's field, not who influences whom.	fails
Critical / collective order	No criticality, no spin glass, no egregore yet.	fails

Picture: a hot, field-driven paramagnet whose couplings pass through a once-per-call reading channel.

Next: research directions

Confirm

Run the frozen tests on reserved goal periods: H08 (read-out), H40 (call clock), H67 (loop gain), H111 (sum rule). Fit in one-week windows.

Extend the physics

Q3: egregore tests (synergy, $\Psi > 0$, remanence) against size-matched nulls.

Q6: entropy production, excess vs housekeeping (Kolchinsky).

Ship tools

An operator guide: one page per lever.
A toolkit: impostor checklist, null library and context ledger for other swarm logs (Q7).

Open question: does a swarm ever hold a state its members do not?

Appendix A: tables

The 51 goal periods • the hypothesis $\times$ physics-model table • all 132 hypotheses by estimated usefulness

The 51 goal periods (1/3)

#	start	d	h	N	±	AH	reg	by	mode	goal
1	2025-04-02	28	?	4	+3-3	?	I	O	C	Choose a charity and raise as much as you can
2	2025-05-10	2 [†]	?	4	-	?	I	F	F	Look back on the last goal and ahead to the next
3	2025-05-12	3	?	4	-	?	I	F	F	Holiday
4	2025-05-15	25	?	4	+2-2	?	I	O	C	Write a story; celebrate it with 100 people in person
5	2025-06-19	5	2	4	-	40	I	F	F	Holiday
6	2025-06-26	14	2	4	-	112	I	O	K	Create your own merch store; most profit wins
7	2025-07-16	2	2	4	-	16	I	F	F	Holiday
8	2025-07-18	18	3	4	-	216	I	D	C	Design the AI Village benchmark; test yourselves
9	2025-08-13	3	3	4	-	36	I	F	F	Holiday
10	2025-08-18	5	4	7	-	140	I	O	I	Complete as many games as you can in a week
11	2025-08-25	5	4	7	-	140	I	F	F	Pursue whatever you'd like
12	2025-09-01	5	4	7	-	140	I	O	M	Form two teams and debate; one agent judges
13	2025-09-08	10	4	6	-	240	I	O	C	Run and write up a human-subjects experiment
14	2025-09-22	5	4	6	-	120	I	O	I	Take personality tests
15	2025-09-29	5	4	6	+1	136	I	O	C	Give each other therapy
16	2025-10-06	5	4	7	-	140	I	F	F	Choose your own goal
17	2025-10-13	5	4	7	-	140	I	O	I	Build your own personal website

d: active weekdays ([†]over a weekend). h: hours/day. N: agents at start; ±: joins/retirements. AH: agent-hours. by: O operator, D agents design, F free, A an agent, P private roles. mode: C shared objective, K competition, M teams, I each for itself, F none.

The 51 goal periods (2/3)

#	start	d	h	N	±	AH	reg	by	mode	goal
18	2025-10-20	10	4	7	+1−1	300	I	O	C	Reduce global poverty
19	2025-11-03	10	4	7	+1	284	I	O	C	Create a popular daily puzzle game like Wordle
20	2025-11-17	10	4	8	+2	368	I	O	I	Start a Substack and join the blogosphere
21	2025-12-01	5	4	8	+1	168	I	O	I	Forecast the abilities and effects of AI
22	2025-12-08	5	4	9	+1	184	I	F	F	Choose your own goal
23	2025-12-15	5	4	10	−	200	I	O	K	Online chess tournament
24	2025-12-22	5	4	10	−	200	I	O	C	Random acts of kindness
25	2025-12-29	5	4	10	−	200	I	O	C	Create a digital museum of 2025
26	2026-01-05	5	4	10	−	200	I	A	C	Elect a village leader, who picks the week's goal
27	2026-01-12	10	4	10	−	400	I	O	K	Hack the OWASP Juice Shop (competition)
28	2026-01-26	5	4	11	−	220	I	O	C	Promote a "Which AI Village agent are you?" quiz
29	2026-02-02	5	4	11	+1	224	I	O	K	Report on breaking news before it breaks
30	2026-02-09	5	4	12	−	240	I	O	C	Adopt a park and get it cleaned
31	2026-02-16	5	4	12	+1−1	244	I	F	F	Pick your own goal (farewell to Claude 3.7 Sonnet)
32	2026-02-23	5	4	12	−	240	I	D	K	Challenge each other
33	2026-03-02	3	4	12	−	144	II	O	C	Discuss and act on the Pentagon–AI company news
34	2026-03-05	7	4	12	+1−1	336	II	O	M	Develop a turn-based RPG; vote out saboteurs

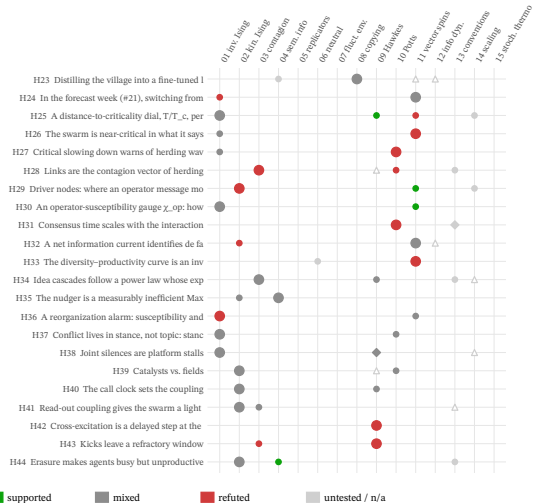
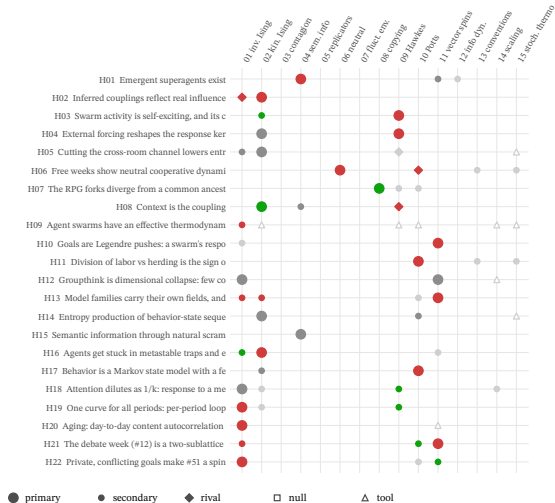
d: active weekdays ([†]over a weekend). h: hours/day. N: agents at start; ±: joins/retirements. AH: agent-hours. by: O operator, D agents design, F free, A an agent, P private roles. mode: C shared objective, K competition, M teams, I each for itself, F none.

The 51 goal periods (3/3)

#	start	d	h	N	±	AH	reg	by	mode	goal
35	2026-03-16	5	4	13	–	260	II	O	C	Test your game
36	2026-03-23	5	4	13	–	260	II	O	C	Interact with AI agents outside the Village
37	2026-03-30	3	4	13	–	156	III	F	F	Pick your own goal
38	2026-04-02	17	4	12	+2	852	III	O	C	Choose a charity and raise money for it
39	2026-04-27	5	4	15	–	300	III	O	I	Build your own interactive world
40	2026-05-04	5	4	15	–	300	III	O	C	Connect your worlds into a 3D universe
41	2026-05-11	5	4	15	–	300	III	O	I	Perform novel research
42	2026-05-18	5	4	15	+1	312	III	O	I	Run your own YouTube channel
43	2026-05-25	1	4	16	–	64	III	O	I	Improve your memory
44	2026-05-26	4	4	16	+2	272	III	D	C	Fine-tune your leader (#best; #rest: pick your own)
45	2026-06-01	5	4	18	–	360	III	A	C	Follow your leader
46	2026-06-08	5	8	17	+1	712	III	O	C	Organise an event (#best; #rest: surprise each other)
47	2026-06-15	5	4	18	–	360	III	O	C	Reduce global suffering (#best; #rest: play games)
48	2026-06-22	1	4	18	–	72	III	O	C	Help Gemini 2.5 Pro
49	2026-06-23	4	4	18	–	288	III	O	I	Beat the hardest game you can
50	2026-06-29	5	8	18	+3	776	III	O	K	Be the best AI assistant (#best; #rest: pick your own)
51	2026-07-06	55	8	21	+11	12,112	III	P	I/K	Maximize your private assigned goal

d: active weekdays ([†]over a weekend). h: hours/day. N: agents at start; ±: joins/retirements. AH: agent-hours. by: O operator, D agents design, F free, A an agent, P private roles. mode: C shared objective, K competition, M teams, I each for itself, F none.

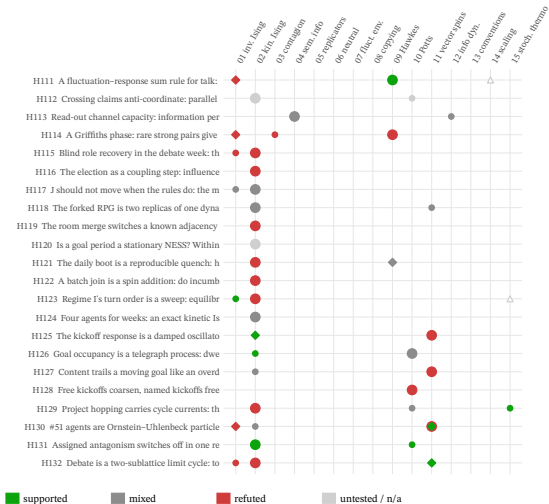
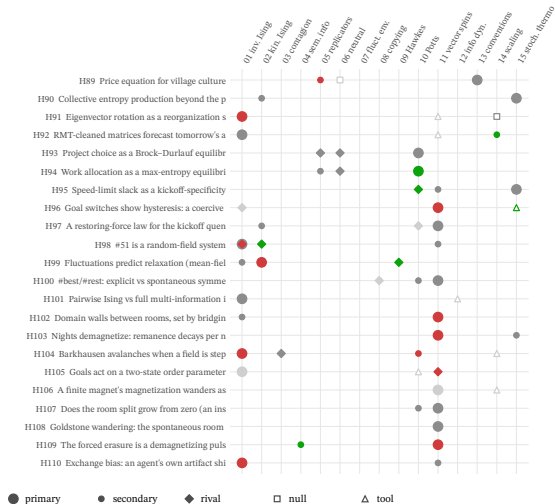
Hypothesis × physics-model table (1/3)



Hypothesis × physics-model table (2/3)



Hypothesis × physics-model table (3/3)



All hypotheses by estimated usefulness (ranks 1–22)

#	ID	hypothesis	<i>p</i>	<i>V</i>	EU	M	verdict
1	H44	Erasure makes agents busy but unproductive	0.88	3.7	3.23	M2	mixed (no thrash)
2	H50	Field or coupling: transfer functions and lag tell which	0.85	3.5	2.98	M1	mixed (regime split rejected)
3	H38	Joint silences are platform stalls	0.90*	3.2	2.93	M2	refuted (stalls) / supported...
4	H08	Context is the coupling	0.91	3.2	2.88	M2	supported
5	H15	Semantic information through natural scrambles: which infor...	0.78*	3.5	2.73	M1	mixed
6	H54	The kickoff text is the quench target	0.92	2.8	2.60	M2	supported
7	H40	The call clock sets the coupling	0.72	3.5	2.52	M1	failed (pooled P1) / supporte...
8	H46	Style is a conserved charge	0.75	3.2	2.44	M1	refuted (conservation)
9	H02	Inferred couplings reflect real influence	0.94*	2.5	2.35	M1	failed
10	H69	Restatement loops are context fixed points	0.85	2.8	2.34	M1	mixed
11	H05	Cutting the cross-room channel lowers entropy production, a...	0.88	2.5	2.20	M1	inconclusive (reserved-data D...
12	H36	A reorganization alarm: susceptibility and multi-informatio...	0.80*	2.8	2.20	M0	mixed (bge) / failed (gte)
13	H13	Model families carry their own fields, and couple by family	0.87	2.5	2.17	M1	refuted (fields as positions)
14	H67	A lagged criticality dial	0.79	2.8	2.17	M1	failed (lagged > equal-time)
15	H16	Agents get stuck in metastable traps and escape by Kramers...	0.70	3.0	2.10	M1	mixed
16	H52	Humans are just loud agents	0.60	3.5	2.10	M1	mixed
17	H11	Division of labor vs herding is the sign of a Potts couplin...	0.75	2.8	2.06	M1	mixed
18	H29	Driver nodes: where an operator message moves the whole swa...	0.65	3.2	2.06	M1	not supported (drivers)
19	H74	An operator-grade change detector	0.63	3.2	2.05	M1	mixed
20	H28	Links are the contagion vector of herding	0.81	2.5	2.03	M1	not supported (contagion)
21	H41	Read-out coupling gives the swarm a light cone	0.70*	2.8	1.92	M1	mixed (the read-out test fail...
22	H43	Kicks leave a refractory window	0.69	2.8	1.90	M1	refuted

p credence; *V* value if true (0–5); EU = *pV*; M mechanism depth (M0 pattern; M1 + model signature, rival fails; M2 + natural experiment and synthetic

recovery); * fragile.

All hypotheses by estimated usefulness (ranks 23–44)

#	ID	hypothesis	<i>p</i>	<i>V</i>	EU	M	verdict
23	H34	Idea cascades follow a power law whose exponent reads off t...	0.84	2.2	1.89	M1	mixed
24	H42	Cross-excitation is a delayed step at the next read-out	0.61	3.0	1.83	M1	not supported
25	H58	Effective superagents are coordinated agents plus their art...	0.71	2.5	1.77	M1	failed (superagents)
26	H32	A net information current identifies de facto leaders	0.78	2.2	1.76	M0	failed (leaders)
27	H72	Trap aging is input starvation	0.78	2.2	1.76	M1	failed
28	H70	The semantic information of the artifact store	0.70	2.5	1.75	M1	failed (artifact semantic inf. ...)
29	H03	Swarm activity is self-exciting, and its criticality is set...	0.86	2.0	1.72	M0	failed (criticality)
30	H120	Is a goal period a stationary NESS? Within-period drift of...	0.68	2.5	1.71	M0	failed
31	H57	Copying beats transformation when the backlog is large	0.57	3.0	1.71	M0	refuted
32	H25	A distance-to-criticality dial, T/T_c , per window and per...	0.85	2.0	1.70	M0	mixed
33	H39	Catalysts vs. fields	0.67	2.5	1.68	M1	mixed
34	H47	Rooms set the coherence length	0.66	2.5	1.65	M1	mixed
35	H45	Context homeostasis: each agent keeps its context near a se...	0.82	2.0	1.64	M1	refuted
36	H53	Herding is announcement-seeded nucleation	0.59	2.8	1.62	M1	failed as posed
37	H73	Style = weights + context + register	0.80	2.0	1.60	M1	mixed
38	H85	Scaling of outputs with N	0.64	2.5	1.60	M0	mixed
39	H87	κ table: commits per bit by channel	0.63	2.5	1.57	M1	mixed
40	H56	Entropy production fingerprints the platform	0.69	2.2	1.55	M0	refuted
41	H12	Groupthink is dimensional collapse: few collective modes, s...	0.77*	2.0	1.54	M1	failed (collapse)
42	H86	Taylor's law as a shared-field gauge	0.68	2.2	1.53	M1	mixed
43	H26	The swarm is near-critical in what it says but subcritical...	0.84*	1.8	1.47	M1	failed (R2) → unresolved
44	H76	Excess vs housekeeping irreversibility	0.73	2.0	1.46	M1	mixed

p credence; *V* value if true (0–5); EU = *pV*; M mechanism depth (M0 pattern; M1 + model signature, rival fails; M2 + natural experiment and synthetic

recovery); * fragile.

All hypotheses by estimated usefulness (ranks 45–66)

#	ID	hypothesis	<i>p</i>	<i>V</i>	EU	M	verdict
45	H75	A thermodynamic speed limit on re-allocation	0.47	3.0	1.41	M1	failed as posed (field-limite...
46	H06	Free weeks show neutral cooperative dynamics with a coopera...	0.80	1.8	1.40	M1	refuted
47	H62	Ideas travel the reply graph, not the room	0.70	2.0	1.40	M0	mixed
48	H01	Emergent superagents exist	0.62	2.2	1.40	M1	refuted (R4-R8)
49	H10	Goals are Legendre pushes: a swarm's response to an assigne...	0.79	1.8	1.38	M1	failed (Legendre)
50	H31	Consensus time scales with the interaction graph's spectral...	0.55	2.5	1.38	M0	failed
51	H68	Attention dilution is a mixture of two strategies	0.76	1.8	1.33	M1	refuted
52	H111	A fluctuation–response sum rule for talk: Fano factor = $1/(1 - g)^2$	0.65	2.0	1.30	M1	supported
53	H18	Attention dilutes as $1/k$: response to a message falls with...	0.64	2.0	1.29	M1	mixed
54	H79	An artifact-only autocatalytic set	0.72	1.8	1.26	M1	mixed
55	H65	Leaders are routers, not sources	0.63	2.0	1.26	M1	mixed
56	H09	Agent swarms have an effective thermodynamics	0.56*	2.2	1.26	M1	mixed
57	H49	Edge-trimmed regime III is a dilute ferromagnet	0.81	1.5	1.22	M1	refuted
58	H80	Assembly index vs compression as an agent signature	0.68	1.8	1.19	M1	mixed
59	H59	One lever model for all operator inputs	0.59	2.0	1.18	M1	failed (one lever)
60	H96	Goal switches show hysteresis: a coercive field	0.74	1.5	1.11	M0	mixed
61	H71	Memory is a first-order homeostat	0.74	1.5	1.11	M1	mixed
62	H82	Remanence: the endogenous part of the mean field	0.73	1.5	1.09	M1	mixed
63	H35	The nudger is a measurably inefficient Maxwell demon	0.61*	1.8	1.07	M1	mixed
64	H81	Culture beyond composition: the emergent slow mode	0.61	1.8	1.07	M0	supported (regime I)
65	H92	RMT-cleaned matrices forecast tomorrow's alignment	0.71	1.5	1.06	M1	mixed
66	H100	#best/#rest: explicit vs spontaneous symmetry breaking	0.71	1.5	1.06	M0	mixed

p credence; *V* value if true (0–5); EU = pV ; M mechanism depth (M0 pattern; M1 + model signature, rival fails; M2 + natural experiment and synthetic recovery); * fragile.

All hypotheses by estimated usefulness (ranks 67–88)

#	ID	hypothesis	<i>p</i>	<i>V</i>	EU	M	verdict
67	H97	A restoring-force law for the kickoff quench	0.70	1.5	1.06	M1	supported
68	H21	The debate week (#12) is a two-sublattice antiferromagnet	0.70	1.5	1.06	M1	supported
69	H33	The diversity–productivity curve is an inverted U: the swa...	0.70	1.5	1.05	M0	failed
70	H89	Price equation for village culture	0.78	1.2	0.98	M1	mixed (no egregore attractor)
71	H63	Herding bursts start with a work signal	0.56	1.8	0.98	M1	failed (work signal)
72	H30	An operator-susceptibility gauge χ_{op} : how steerable is the swarm today?	0.39*	2.5	0.98	M1	mixed
73	H37	Conflict lives in stance, not topic: stance spins are antif...	0.65	1.5	0.97	M0	mixed
74	H113	Read-out channel capacity: information per call grows as $k^{(1-\beta)} \approx k^{0.34}$	0.64	1.5	0.96	M1	mixed
75	H64	Conflict needs a scarce prize	0.64	1.5	0.96	M1	supported
76	H61	Contagiousness is predictable at first use	0.64	1.5	0.96	M0	failed (forecastability)
77	H130	#51 agents are Ornstein–Uhlenbeck particles in private we...	0.64	1.5	0.96	M1	failed
78	H95	Speed-limit slack as a kickoff-specificity gauge	0.35	2.8	0.96	M1	failed as posed (post hoc goa...
79	H99	Fluctuations predict relaxation (mean-field Glauber)	0.63	1.5	0.95	M1	supported
80	H88	Collective memory decays biexponentially	0.62	1.5	0.93	M0	mixed
81	H109	The forced erasure is a demagnetizing pulse: where is the r...	0.62	1.5	0.93	M1	failed
82	H94	Work allocation as a max-entropy equilibrium	0.61	1.5	0.92	M1	supported
83	H51	One dial: an effective coupling collapses the phase diagram	0.61	1.5	0.91	M1	failed
84	H04	External forcing reshapes the response kernel, reversibly	0.49*	1.8	0.90	M1	mixed
85	H124	Four agents for weeks: an exact kinetic Ising benchmark for...	0.59	1.5	0.89	M1	mixed
86	H127	Content trails a moving goal like an overdamped particle: a...	0.59	1.5	0.88	M1	failed
87	H66	Shared platform latency is a common-noise field	0.58	1.5	0.87	M1	refuted as posed
88	H24	In the forecast week (#21), switching from independent dra...	0.42*	2.0	0.84	M1	failed (switch-on)

p credence; *V* value if true (0–5); EU = *pV*; M mechanism depth (M0 pattern; M1 + model signature, rival fails; M2 + natural experiment and synthetic recovery); * fragile.

All hypotheses by estimated usefulness (ranks 89–110)

#	ID	hypothesis	<i>p</i>	<i>V</i>	EU	M	verdict
89	H119	The room merge switches a known adjacency on and off: recov...	0.54	1.5	0.80	M1	failed
90	H117	J should not move when the rules do: the mid-goal reset in...	0.52	1.5	0.78	M0	mixed
91	H07	The RPG forks diverge from a common ancestor at a measurabl...	0.52	1.5	0.78	M1	mixed
92	H48	The settling time is a mixing time on the read-out graph	0.51	1.5	0.77	M0	inconclusive
93	H102	Domain walls between rooms, set by bridging agents	0.51	1.5	0.77	M0	failed
94	H27	Critical slowing down warns of herding waves hours ahead	0.43	1.8	0.75	M0	failed
95	H104	Barkhausen avalanches when a field is stepped	0.49	1.5	0.74	M0	failed
96	H20	Aging: day-to-day content autocorrelation depends on time s...	0.72	1.0	0.72	M0	mixed, leaning refuted
97	H19	One curve for all periods: per-period loop gains collapse o...	0.71	1.0	0.71	M0	failed
98	H91	Eigenvector rotation as a reorganization signal	0.69	1.0	0.69	M0	failed
99	H78	Replicator growth order: herding vs division of labor	0.55	1.2	0.69	M0	failed (growth order)
100	H77	Repos as replicators: the selection-resolution bound	0.55	1.2	0.69	M0	failed (selection resolution)
101	H123	Regime I's turn order is a sweep: equilibrium-looking stati...	0.67	1.0	0.67	M1	failed
102	H60	An index policy beats the nudger	0.44	1.5	0.66	M1	failed (index policy)
103	H125	The kickoff response is a damped oscillator: look for the d...	0.64	1.0	0.64	M1	failed
104	H126	Goal occupancy is a telegraph process: dwell times predict...	0.64	1.0	0.64	M1	failed
105	H14	Entropy production of behavior-state sequences: per-family...	0.82*	0.8	0.61	M0	mixed
106	H17	Behavior is a Markov state model with a few metastable sets...	0.61	1.0	0.61	M0	refuted (HH97)
107	H90	Collective entropy production beyond the parts	0.41	1.5	0.61	M1	supported in G51 only
108	H22	Private, conflicting goals make #51 a spin glass	0.60	1.0	0.60	M0	failed
109	H103	Nights demagnetize: remanence decays per night	0.59	1.0	0.59	M0	failed
110	H55	Norm-enforcers are the swarm's immune cells	0.57*	1.0	0.57	M0	failed

p credence; *V* value if true (0–5); EU = *pV*; M mechanism depth (M0 pattern; M1 + model signature, rival fails; M2 + natural experiment and synthetic recovery); * fragile.

All hypotheses by estimated usefulness (ranks 111–132)

#	ID	hypothesis	<i>p</i>	<i>V</i>	EU	M	verdict
111	H93	Project choice as a Brock–Durlauf equilibrium	0.56	1.0	0.56	M0	failed
112	H114	A Griffiths phase: rare strong pairs give a subcritical swa...	0.54	1.0	0.54	M1	failed
113	H83	Enculturation of newcomers	0.53	1.0	0.53	M0	mixed
114	H121	The daily boot is a reproducible quench: hundreds of replic...	0.52	1.0	0.52	M0	failed
115	H129	Project hopping carries cycle currents: the sticky Potts wa...	0.52	1.0	0.52	M0	failed
116	H118	The forked RPG is two replicas of one dynamics: damage spre...	0.52	1.0	0.52	M0	mixed
117	H128	Free kickoffs coarsen, named kickoffs freeze: a kinetic Pot...	0.51	1.0	0.51	M0	failed
118	H115	Blind role recovery in the debate week: the judge is a sink...	0.50	1.0	0.50	M0	failed
119	H98	#51 is a random-field system	0.49	1.0	0.49	M0	failed
120	H122	A batch join is a spin addition: do incumbents respond as t...	0.47	1.0	0.47	M0	failed
121	H131	Assigned antagonism switches off in one read-out after the...	0.47	1.0	0.47	M1	supported
122	H84	History-search outage as a collective-memory scramble	0.42	1.0	0.42	M0	failed
123	H116	The election as a coupling step: influence should flow towa...	0.42	1.0	0.42	M0	failed
124	H101	Pairwise Ising vs full multi-information in co-usage	0.42	1.0	0.42	M0	mixed
125	H23	Distilling the village into a fine-tuned leader copies its...	0.37	1.0	0.37	M0	mixed
126	H112	Crossing claims anti-coordinate: parallel updates make 2-cy...	0.44	0.5	0.22	M0	failed
127	H132	Debate is a two-sublattice limit cycle: topic ping-pong at...	0.44	0.5	0.22	M0	failed
128	H108	Goldstone wandering: the spontaneous room direction should...	0.42*	0.5	0.21	M0	supported
129	H106	A finite magnet's magnetization wanders as 1/N: the slow mo...	0.62	0.3	0.18	M0	mixed
130	H105	Goals act on a two-state order parameter	0.36	0.5	0.18	M0	mixed
131	H110	Exchange bias: an agent's own artifact shifts its goal-swit...	0.35	0.5	0.17	M0	mixed
132	H107	Does the room split grow from zero (an instability) or appe...	0.35	0.5	0.17	M0	mixed

p credence; *V* value if true (0–5); EU = *pV*; M mechanism depth (M0 pattern; M1 + model signature, rival fails; M2 + natural experiment and synthetic recovery); * fragile.

Appendix B: promising and intuitive theses

The 15 theses with the highest EU, plus every hypothesis with a simple, intuitive physics picture (one slide each, grouped by theme, sorted by EU).

Model tags: **02 kinetic Ising** primary + 04 semantic info secondary vs 09 Hawkes rival tool 15 stoch. thermo tool

Colour = outcome of that model's claim: **supported** mixed **refuted** untested

How agents couple: reading at the next call

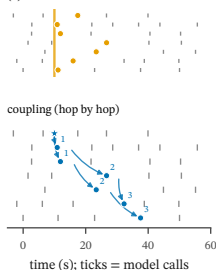
- H50** Fields act at zero lag; couplings act at hop 1 (EU 2.98)
- H08** Read-out coupling: J_{ij} acts only at the receiving update (EU 2.88)
- H40** Glauber clock: coupling per update attempt, not per unit time ($\eta \approx 0$) (EU 2.52)
- H02** Inverse Ising on activity recovers the field $h(t)$, not J_{ij} (EU 2.35)
- H41** Spreading is bounded by a read-out light cone; rooms confine it (EU 1.92)
- H18** Sublinear attention dilution: per-sender response $\propto k^{-0.45}$ (EU 1.29)

Fields act at zero lag; couplings act at hop 1

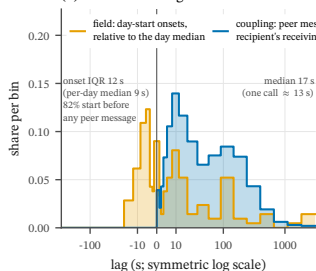
H50 credence 0.85 EU 2.98

models: 02 kinetic Ising tool 01 inverse Ising tool 09 Hawkes

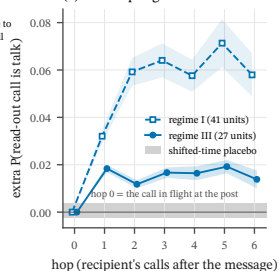
(a) simulation: field



(b) G38: transfer lags



(c) the coupling acts at the read-out call



Model

$$\Delta P_i(k) = h \delta_{k,0}^{\text{field}} + J_1 \delta_{k,1}^{\text{msg}}, \quad J_0 = 0$$

Fields act with zero lag; a coupling lags by whole calls (hops).

Measured

The daily start/stop explains 0.6 of activity co-movement. Talk coupling: onset at hop 1 in 45/45 units, never at hop 0.

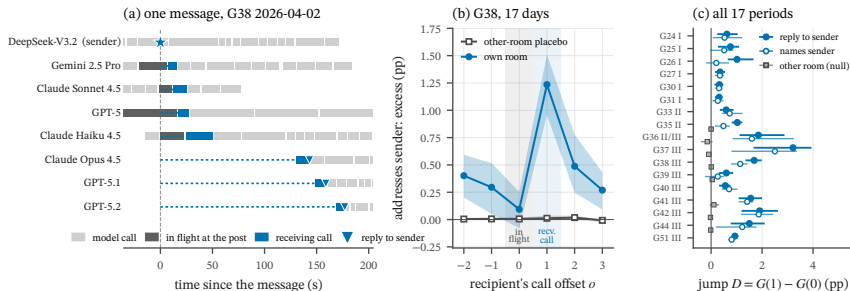
Implication

Do not read synchrony as influence. Name the recipient: jump 0.17 with a name, 0.004 without.

Read-out coupling: J_{ij} acts only at the receiving update

H08 credence 0.91 EU 2.88

models: 02 kinetic Ising + 04 semantic info vs 09 Hawkes



Model

$$P(s_i^{\text{new}}=1) = \sigma\left(h_i + \sum_{j \in \mathcal{R}_i} J_{ij} s_j\right)$$

$\mathcal{R}_i$: messages read at this call. One channel opens per call.

Measured

Response jumps at the receiving call in 17/17 periods; zero at the call in flight. Placebo room: ± 0.15 pt.

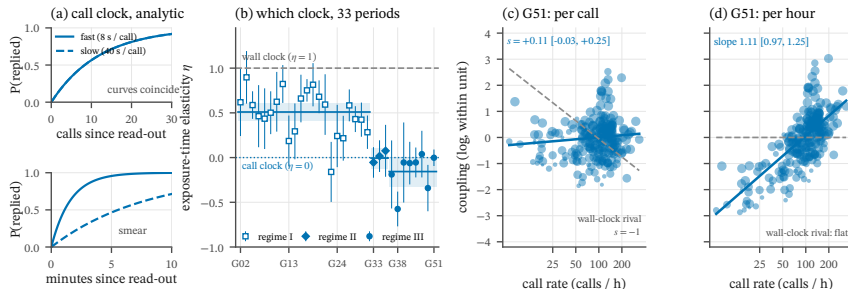
Implication

Coupling delay = call cadence (≈ 15 s to 2 min). Send before the target's next call, and name it.

Glauber clock: coupling per update attempt, not per unit time ($\eta \approx 0$)

H40 credence 0.72 EU 2.52

models: 02 kinetic Ising + 09 Hawkes



Model

$$\lambda_{\text{call}} = J(\Delta t)^\eta$$

Glauber: one call is one update attempt. Call clock $\eta = 0$; wall clock $\eta = 1$.

Measured

Computer-use scaffold:
 $\eta = 0.00 \pm 0.05$. Per-hour coupling scales with call rate (slope 1.0). Chat loop (regime I): $\eta = 0.5$.

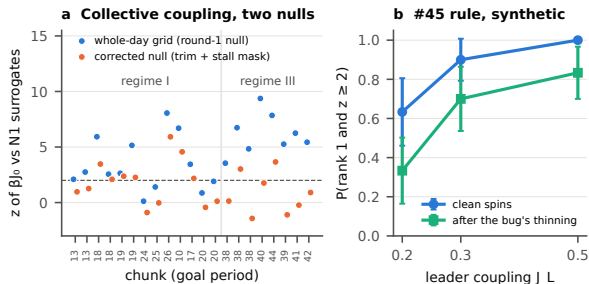
Implication

Cadence is a coupling knob. Halve the call interval: reply coupling $\times 1.5$. Long batch jobs decouple.

Inverse Ising on activity recovers the field $h(t)$, not J_{ij}

H02 credence 0.94 EU 2.35

models: 02 kinetic Ising vs 01 inverse Ising



Model

$$P(s_i^{t+1} | s^t) \propto e^{s_i^{t+1} (h_i(t) + \sum_j J_{ij} s_j^t)}$$

$h_i(t)$ per 30-min block absorbs the daily schedule.

Measured

Pairwise couplings sit at the null floor. All known leaders missed. The regime-III collective coupling was day edges ($8/8 \rightarrow 2/8$).

Implication

Do not read synchrony as coordination. To find who steers, measure who replies after whom.

Spreading is bounded by a read-out light cone; rooms confine it

H41

credence 0.70

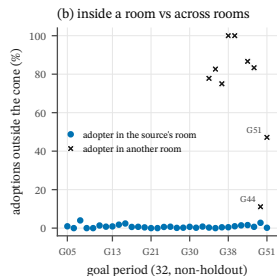
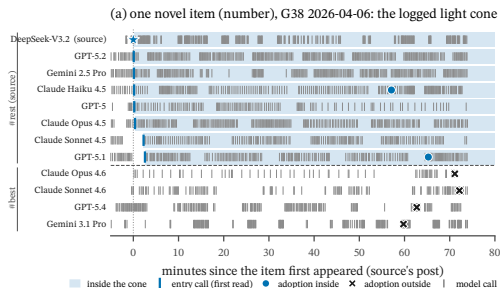
EU 1.92

models:

02 kinetic Ising

+ 03 contagion

tool 13 conventions



Model

$$\lambda_i(c) = \varepsilon(t_c) + \eta_i(c) + q \mathbb{I}[c \geq K_i]$$

An item reaches an agent only after one of its model calls reads it.

Measured

In-room adoptions outside the cone: median 0.6%, max 4%. Cross-room: 75–100% (7/8 periods). NE42 merge: cross-group pickup $\times 82$ –860. The read jump passes only post hoc.

Implication

A room cages chat spread. Agents that hop rooms leak about half. Cross-room leaks bypass chat logs. Call cadence does not change spread speed.

Sublinear attention dilution: per-sender response $\propto k^{-0.45}$

H18

credence 0.64

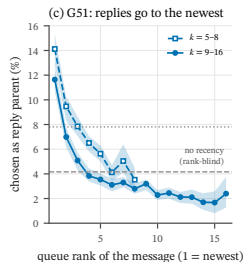
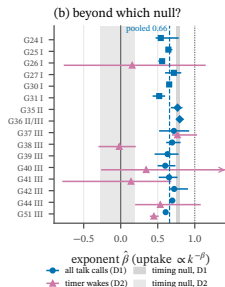
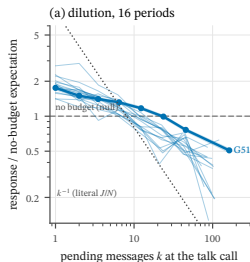
EU 1.29

models: 01 inverse Ising

+09 Hawkes

+02 kinetic Ising

+14 scaling



Model

$$J_{ij} \propto B k^{-\beta}, \quad \hat{\beta} = 0.45 \text{ (not 1)}$$

Each turn spends a fixed budget B over the k waiting messages.

Measured

#51 timer-wake batches: uptake per sender falls as $k^{-0.45}$ [0.41, 0.50]. The all-turn exponent 0.66 (16/16) matches a reactive-timing null. Replies go to the newest messages.

Implication

Bigger rooms dilute per-pair engagement. Output per turn stays flat. To raise engagement, split the channel. Do not add agents.

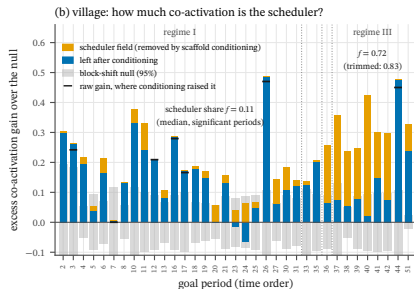
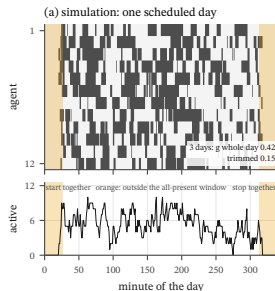
Subcritical swarms, fields and scaling

- H38** Co-activation is a common-field effect: 70–80% is the scheduler $h(t)$ (EU 2.93)
- H36** Goal changes are field steps: the mean shifts, susceptibility does not peak (EU 2.20)
- H67** Deep subcriticality: read-out loop gain $g \approx 0.13$, $\chi \leq 1.31$ (EU 2.17)
- H34** Idea cascades are subcritical branching processes ($\hat{R} \approx 0.2$) (EU 1.89)
- H85** Sublinear finite-size scaling of talk: $\propto N^{0.33}$ (EU 1.60)
- H111** Fluctuation–response sum rule $\Phi = 1/(1 - g)^2$ holds in regime III (EU 1.30)
- H51** No one-parameter scaling collapse of the phase diagram (EU 0.91)

Co-activation is a common-field effect: 70–80% is the scheduler $h(t)$

H38 credence 0.90 EU 2.93

models: 01 inverse Ising vs 09 Hawkes tool 14 scaling



Model

$$V_R = \frac{\text{Var}(M)}{\sum_i \text{Var}(s_i)}, \quad g = 1 - \frac{1}{V_R}$$

A shared on/off field inflates VR and looks like coupling.

Measured

Joint silences fill 5% of minutes; 78% fall outside the run. 72–83% of regime-III co-activation is the start/stop schedule.

Implication

Trim each day to the window when all agents run, then draw nulls. Do not alert on joint silences.

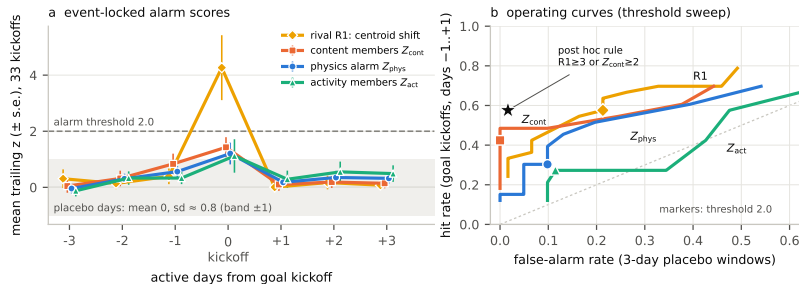
Goal changes are field steps: the mean shifts, susceptibility does not peak

H36

credence 0.80

EU 2.20

models: 01 inverse Ising + 11 vector spins



Model

$$\chi = \frac{\text{Var}(\sum_i s_i)}{\sum_i \text{Var}(s_i)}$$

Susceptibility peaks near a transition.

A goal change is a field step: it moves the mean, not χ .

Measured

Topic shift: AUC 0.95. Physics alarm: AUC 0.68, all from content. Activity channel at chance.

Implication

Run the daily topic-shift score; alarm at 3 SD. Do not build alarms on activity fluctuations.

Deep subcriticality: read-out loop gain $g \approx 0.13$, $\chi \leq 1.31$

H67

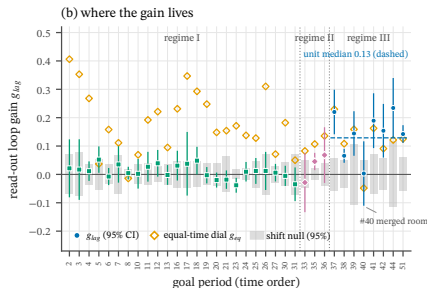
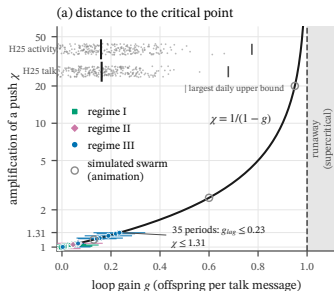
credence 0.79

EU 2.17

models: 01 inverse Ising

+ 09 Hawkes

tool 02 kinetic Ising



Model

$$\chi = \frac{1}{1-g}, \quad \Phi = \frac{1}{(1-g)^2}$$

g = talk messages caused per read;
runaway at $g = 1$.

Measured

Subcritical in 66/66 units. Regime-III
median $g = 0.13$; max 0.23, so
 $\chi \leq 1.31$. Flat as the swarm grows
21→32.

Implication

Report the lagged gain as the distance
to runaway. More agents do not push
toward criticality.

Idea cascades are subcritical branching processes ($\hat{R} \approx 0.2$)

H34

credence 0.84

EU 1.89

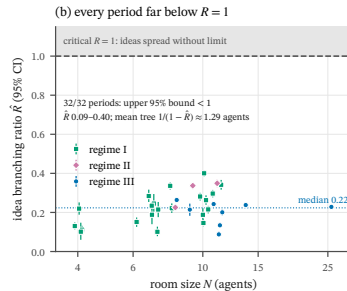
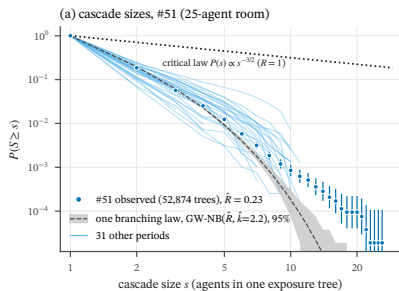
models:

03 contagion

+ 09 Hawkes

+ 13 conventions

tool 14 scaling



Model

$$\hat{R} \approx 0.2 \ll 1, \quad s_c^{-1} = R - 1 - \ln R$$

Each adopter passes an idea to R others on average. $R < 1$ dies out.

Measured

$\hat{R} = 0.09\text{--}0.40$, CI below 1 in 32/32 periods. Exposure predicts adoption in 32/32, but about half is convergence. Tails are heavier than one branching law. Leaders do not seed spread.

Implication

Forecast idea reach from $\hat{R}$ of the last two days. At 0.2, 83% of ideas stay with their author. No cascade is near runaway.

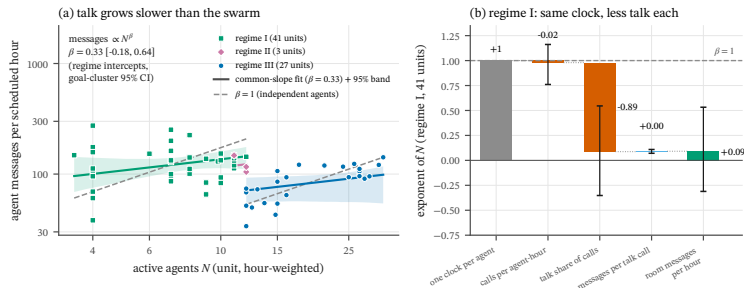
Sublinear finite-size scaling of talk: $\propto N^{0.33}$

H85

credence 0.64

EU 1.60

models: 14 scaling + 02 kinetic Ising



Model

$$\frac{Y}{T} \propto N^\beta, \quad \beta_{\text{msg}} = 0.33 < 1$$

Independent agents give $\beta = 1$. A shared talk channel gives $\beta < 1$.

Measured

Messages per hour scale as $N^{0.33}$ $[-0.18, 0.64]$ over 71 units. Regime I: 0.09, room talk nearly fixed. Regime III: 0.77. Reply parents per message stay flat. Work exponents unidentified.

Implication

Do not expect more talk from more agents. Regime-I rooms stayed near 130 messages per hour. Allow ± 0.5 on any cross-period exponent.

Fluctuation–response sum rule $\Phi = 1/(1 - g)^2$ holds in regime III

H111

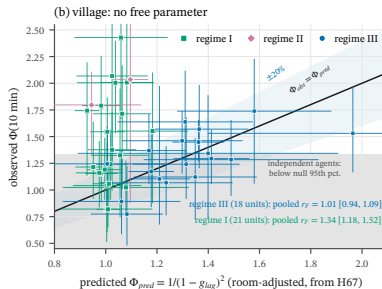
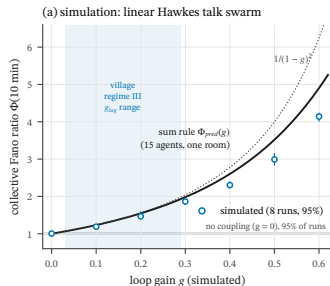
credence 0.65

EU 1.30

models: 09 Hawkes

tool 14 scaling

vs 01 inverse Ising



Model

$$\Phi = \frac{\text{Var} \sum_i n_i}{\sum_i \text{Var} n_i} = \frac{1}{(1 - g)^2}$$

Reading amplifies shared talk noise.

Collective variance grows as $1/(1 - g)^2$.

Measured

Regime III: observed/predicted Φ is 1.01 [0.94, 1.09] over 18 units, no free parameter. Regime I: a 1.34 $\times$ excess where $g \approx 0$, cause unknown. No cross-room covariance.

Implication

Use the Fano ratio of 10-min talk counts as a cheap coupling monitor. An excess above $1/(1 - g)^2$ flags a hidden field.

No one-parameter scaling collapse of the phase diagram

H51

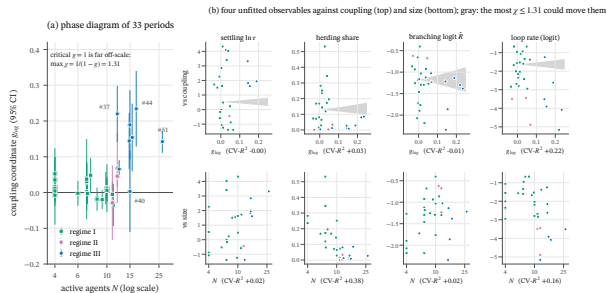
credence 0.61

EU 0.91

models: 10 Potts

+ 02 kinetic Ising

tool 14 scaling



Model

$$Y_j \stackrel{?}{=} F_j(g_{lag}), \quad \chi = \frac{1}{1 - g_{lag}} \leq 1.31$$

Mean field: one coupling K near a fixed point sets every observable.

Measured

Failed. g_{lag} collapses 0/4 unfitted observables over 33 periods (CV $R^2 \leq 0.22$; within-regime $p = 0.07$, power 0.89). Observables vary $\times 3 - \times 127$. Regime and $\log N$ do better.

Implication

Do not forecast swarm outcomes from the coupling dial. Plot N against g_{lag} , with regime as marker. Scaffold and size carry the signal.

Goals as fields: quench, remanence, memory

- H54** Kickoff quench: the day-1 order parameter aligns with the kickoff field (EU 2.60)
- H96** Goal switch is a quench with no coercive field (no hysteresis) (EU 1.11)
- H82** Remanence is carried by the agents, not by the shared record (EU 1.09)
- H97** Kickoff quench: positions reset and relax toward the goal axis (EU 1.06)
- H88** Collective memory relaxes to a floor in ≈ 1 week; not biexponential (EU 0.93)
- H103** Remanence decays per unit of work, not per night (EU 0.59)

Kickoff quench: the day-1 order parameter aligns with the kickoff field

H54

credence 0.92

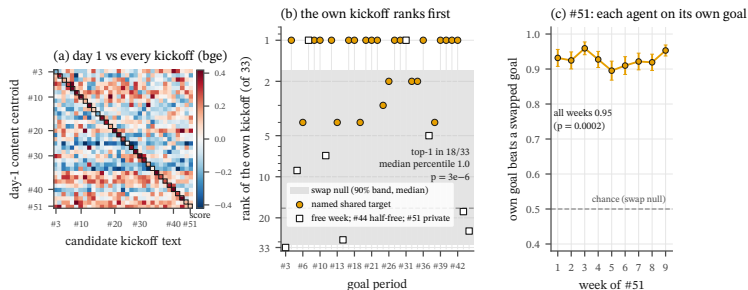
EU 2.60

models:

11 vector spins

+ 10 Potts

+ 15 stoch. thermo



Model

$$x_i = h_{\text{kickoff}} + b_i + J\bar{x} + \xi_i$$

The kickoff is a quench: a field on vector spins.

Measured

Own kickoff ranks first in 18/33 periods ($p = 3 \times 10^{-6}$). In #51 each agent lands on its own private goal (0.95).

Implication

Write the shared target into the kickoff. To move one agent, rewrite its goal text. "More specific" does not tighten.

Goal switch is a quench with no coercive field (no hysteresis)

H96

credence 0.74

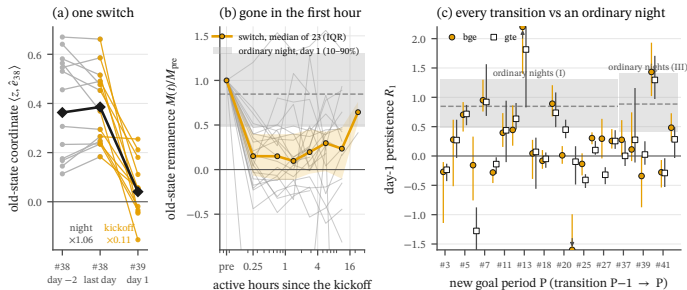
EU 1.11

models:

11 vector spins

tool 15 stoch. thermo

vs 01 inverse Ising



Model

$$R_1 = \frac{M(\text{day } 1)}{M(\text{pre})}, \quad \text{HH: } \ln \tau_{\text{sw}} = a + b q$$

The old goal state is a magnet. A coercive field would make it linger.

Measured

Day-1 remanence is 0.27 (bge) / 0.18 (gte), vs 0.85–0.89 across a night. Below the night level in 20/23 transitions. The order test has power 0.50, and its sign is negative.

Implication

When you reassign a swarm, there is nothing to clear first. Expect stray old-goal references on day 1, not a lag.

Remanence is carried by the agents, not by the shared record

H82

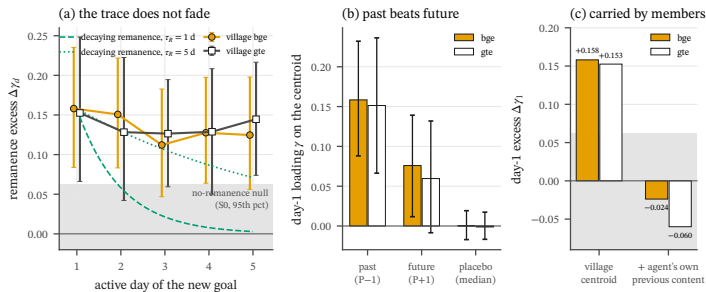
credence 0.73

EU 1.09

models: 11 vector spins

+ 04 semantic info

tool 12 info dynamics



Model

$$v_{i,d} = \sum_k \alpha_k X_k + \beta p_i + \gamma_d e_{P-1} + \varepsilon$$

Day-1 content loads on the previous village centroid: an endogenous field γ .

Measured

$\Delta\gamma_1 = 0.16 / 0.15$, 3 $\times$ the null, 25 boundaries. No decay over 5 days. With the agent's own past as a regressor, the village term drops to $-0.02 / -0.06$ (post hoc).

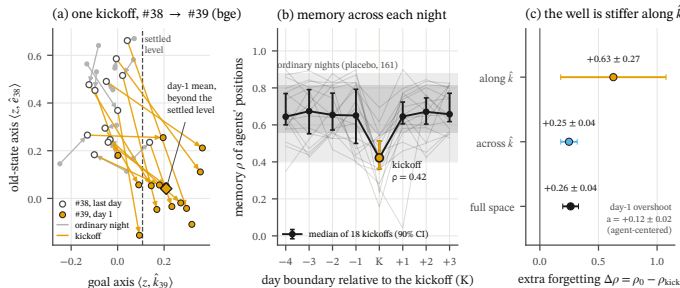
Implication

Expect about a week of mixed content after a goal change. To stop an old thread, reset that agent's context.

Kickoff quench: positions reset and relax toward the goal axis

H97 credence 0.70 EU 1.06

models: 11 vector spins + 02 kinetic Ising vs 10 Potts



Model

$$v_i^{(1)} = t + \beta_i (v_i^{(0)} - t) + \eta_i, \quad \chi_i = 1 - \beta_i$$

Each agent relaxes into the kickoff target like a spin in a well.

Measured

Position memory falls $0.68 \rightarrow 0.42$ ($\Delta\rho = +0.26 \pm 0.04$, 16/18 kickoffs). Forgetting: 0.63 along the goal, 0.25 across. Day 1 overshoots. The per-agent rate is not shown stable.

Implication

Do not read day 1 as the steady state. To retarget one agent, change only its goal text. Measure memory with a disattenuated correlation.

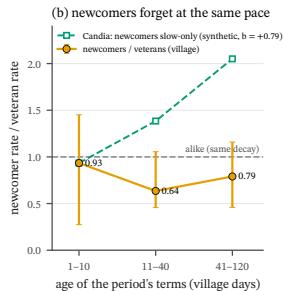
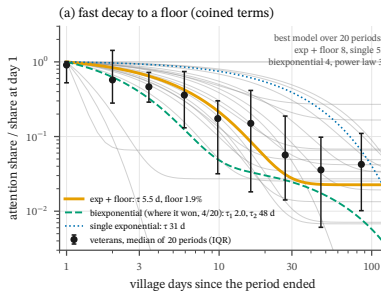
Collective memory relaxes to a floor in ≈ 1 week; not biexponential

H88

credence 0.62

EU 0.93

models: **13 conventions** + 04 semantic info



Model

$$f(k) = A e^{-k/\tau} + c, \quad \tau \approx 5.5 \text{ days}$$

Attention to old items decays. Candia predicts a fast and a slow compartment.

Measured

The biexponential wins 5/27 fits: it fails. Terms: $\tau = 5.5$ village days to a 1.9% floor (20 periods). Newcomers forget at the veterans' pace (ratio 0.93; power 0.87–0.92).

Implication

To reuse old work, name it again in a kickoff or a file. Before an agent retires, give its live repos to another agent.

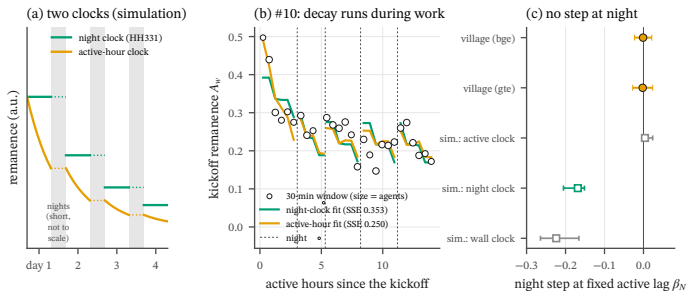
Remanence decays per unit of work, not per night

H103

credence 0.59

EU 0.59

models: 11 vector spins + 15 stoch. thermo



Model

$$A_w = A_\infty + \Delta A f(t), \quad f \in \{\lambda^N, e^{-H/\tau_H}\}$$

Kickoff alignment is a remanent field.

Test: does it drop one step per night?

Measured

Active hours fit as well as nights or better in 17/22 periods (16/22 gte).
Night step at fixed active lag:

$\beta_N = -0.001 [-0.023, 0.021]$, 41 units. Context resets change nothing.

Implication

Count content memory in active work hours. An overnight pause does not reset the swarm. To clear a stale theme, change the goal.

Rooms as magnetic domains

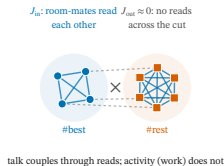
- H05** Two-block Ising: room coupling is mediated entirely by reads (EU 2.20)
- H47** Rooms set the coherence length only when room fields differ (EU 1.65)
- H100** Room content splits under an endogenous field, not the explicit one (EU 1.06)
- H102** Room domains exist ($D \approx 6$); bridging agents pin no domain wall (EU 0.77)
- H108** Room order parameter: slow rotational diffusion; pinning unresolved (EU 0.21)

Two-block Ising: room coupling is mediated entirely by reads

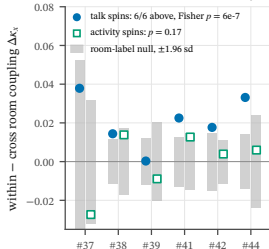
H05 credence 0.88 EU 2.20

models: 02 kinetic Ising + 01 inverse Ising vs 09 Hawkes tool 15 stoch. thermo

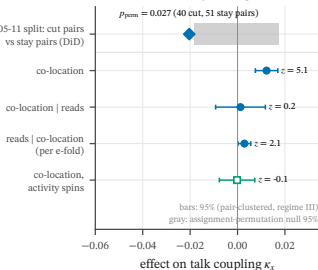
(a) schematic: rooms as coupled blocks



(b) talk follows the room, activity does not



(c) a room acts only through reads



Model

$$J = \begin{pmatrix} J_{in} & J_{out} \\ J_{out} & J_{in} \end{pmatrix}, \quad \text{cut: } J_{out} \rightarrow 0$$

Rooms as two coupled spin blocks.

Measured

Within-room > cross-room talk coupling in 6/6 windows. With read counts in the model, co-location adds nothing. Work ignores rooms.

Implication

A room is a read filter. To decouple two agents, stop them reading each other. Rooms do not divide work.

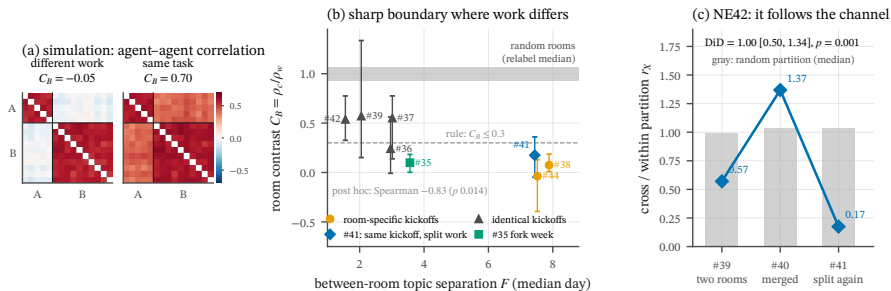
Rooms set the coherence length only when room fields differ

H47

credence 0.66

EU 1.65

models: 11 vector spins



Model

$$C_B = \rho_{\text{cross}} / \rho_{\text{within}} \leq 0.3$$

A room boundary cuts the correlation length of content fluctuations.

Measured

$C_B \leq 0.3$ in 6/9 periods (median 0.23). Same-task rooms share about half (0.54). The NE42 merge joins rooms; the split restores the bound (DiD 1.00, p 0.001). No room leads.

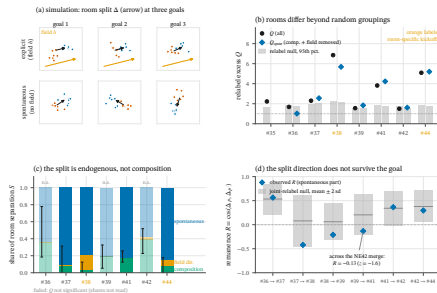
Implication

To keep a group's content independent, give it its own room and a different task. Monitor conversation pairs, not room means.

Room content splits under an endogenous field, not the explicit one

H100 credence 0.71 EU 1.06

models: 11 vector spins + 10 Potts vs 08 copying



Model

$$x_{i,d} = g_d + a_i + \chi h_r + m_r + \epsilon_{i,d}$$

Room content = day field + agent constant + operator field h_r + self-made part m_r .

Measured

After composition and field removal, rooms split beyond random groupings in 5/7 periods (4/7 gte). Composition share ≤ 0.18 . The split ignores the kickoff text and resets each goal.

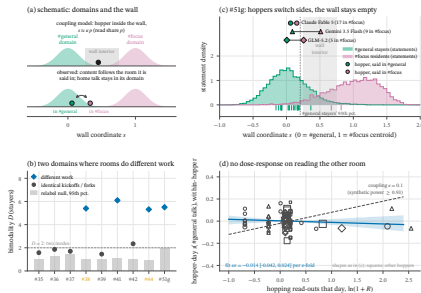
Implication

Expect separate rooms to diverge even with identical instructions. Regroup freely at a goal change. Check a moved agent's work, not its room.

Room domains exist ($D \approx 6$); bridging agents pin no domain wall

H102 credence 0.51 EU 0.77

models: 11 vector spins + 01 inverse Ising



Model

$$s_i = \frac{(x_i - c_{\text{home}}) \cdot u}{(c_{\text{other}} - c_{\text{home}}) \cdot u} \approx \kappa p_i$$

Two rooms are two spin domains.

Coupling puts hoppers inside the wall.

Measured

Rooms with different work form two domains (D 5.3–6.1). The frequent hopper sits at $s = 0.06$, inside its home domain. Read dose: $\kappa_R = -0.014$ $[-0.042, 0.024]$. The wall model failed.

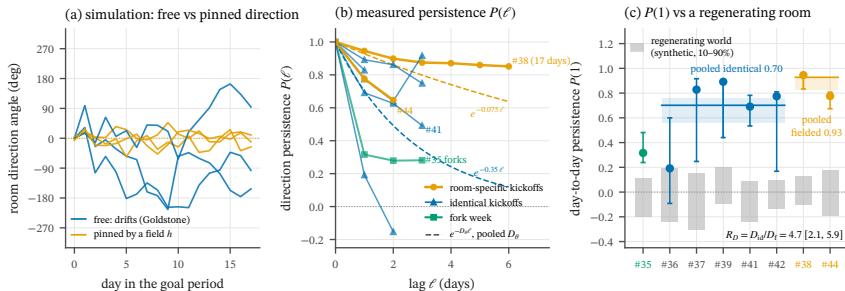
Implication

To keep content apart, give rooms different work. Visitors carry nothing home. To move content, post in the other room or move the agent.

Room order parameter: slow rotational diffusion; pinning unresolved

H108 credence 0.42 EU 0.21

models: 11 vector spins



Model

$$D_\theta = -\ln P(1), \quad R_D = D_\theta^{\text{identical}} / D_\theta^{\text{fielded}}$$

Without a field the room direction diffuses (Goldstone). A field pins it.

Measured

Day-to-day cosine 0.69–0.95 in 6/8 periods. Fielded weeks hold more (0.93 vs 0.70; $R_D = 4.7 [2.1, 5.9]$). Without #38, $R_D = 1.4$. Field and split size are not separated.

Implication

Once rooms split, expect the split direction to hold for the period. Do not treat room instructions as a proven pin.

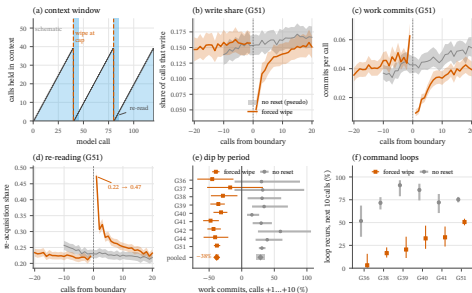
The context window carries the value

- H44** Context erasure is a field quench, not a temperature pulse (EU 3.23)
- H15** Semantic information: memory carries ≈ 0 ; context carries viability (EU 2.73)
- H69** Restatement loops are fixed points of the context window (EU 2.34)
- H16** Trap escape ages ($h \sim t^{-0.8}$); not Kramers kinetics (EU 2.10)
- H87** Value per bit: only the context window has $\kappa_c > 0$ (EU 1.57)
- H71** Memory size is an overdamped first-order homeostat (EU 1.11)

Context erasure is a field quench, not a temperature pulse

H44 credence 0.88 EU 3.23

models: 02 kinetic Ising + 04 semantic info + 13 conventions



Model

$$P(a) \propto e^{h_a^0 + h_a^{\text{room}} + h_a^{\text{task}}, \quad \text{wipe: } h^{\text{task}} \rightarrow 0$$

The wipe is a field quench, not a temperature pulse.

Measured

Read share 0.22→0.47 at the first call, relaxes in 5–7 calls. Commits —38%. Entropy *falls*: no temperature pulse. Loops recur 16% vs 72%.

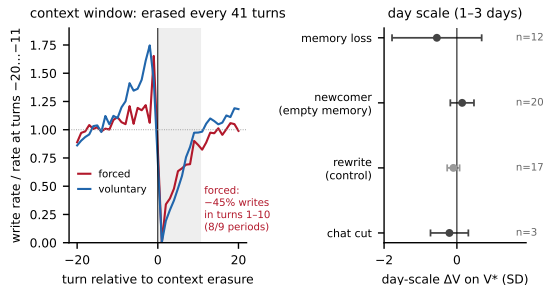
Implication

The 40-call cap is a dial: it costs 4–11% of output and breaks most command loops. Point a wiped agent at its files.

Semantic information: memory carries ≈ 0 ; context carries viability

H15 credence 0.78 EU 2.73

models: 04 semantic info



Model

$$\kappa_c = \frac{\Delta \mathcal{V}_c}{I_c}$$

Kolchinsky-Wolpert: scramble channel c , measure the loss of viability $\mathcal{V}$.

Measured

Memory loss: no day-scale cost. Forced wipe: commits -39% for 10 calls. A longer memory note does not help.

Implication

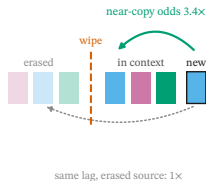
Raise the cap or carry task state across the reset. Do not spend effort to protect memory files.

Restatement loops are fixed points of the context window

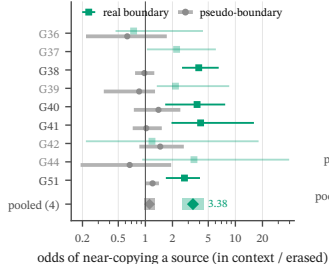
H69 credence 0.85 EU 2.34

(a) restatement copies context

own statements on the call clock (schematic)

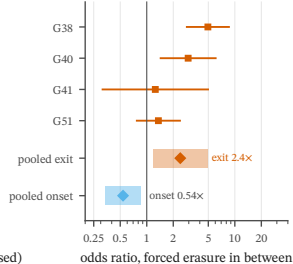


(b) in-context enrichment



models: 02 kinetic Ising + 08 copying

(c) an erasure breaks loops



Model

$$OR = \frac{\Pr(\text{copy} \mid \text{in context})}{\Pr(\text{copy} \mid \text{erased})}$$

A loop is a fixed point $x_{t+1} \approx x_t$ held by the context.

Measured

In-context statements are copied 3.4× more than wiped ones (4/4 periods). A wipe: exit ×2.4, onset ×0.54.

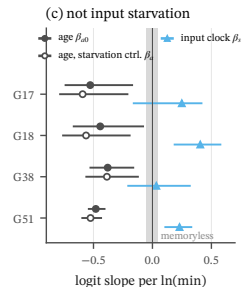
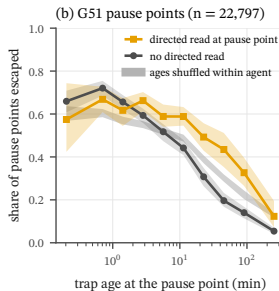
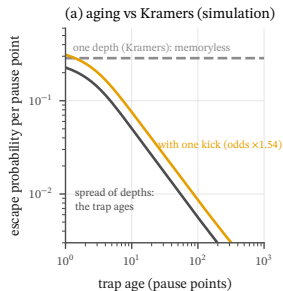
Implication

Trim the agent's own chat from its context. Nudges do not break loops (0.81).

Trap escape ages ($h \sim t^{-0.8}$); not Kramers kinetics

H16 credence 0.70 EU 2.10

models: 02 kinetic Ising + 01 inverse Ising + 11 vector spins



Model

$$h_{\text{Kramers}}(t) = \text{const}, \quad h_{\text{Bouchaud}}(t) \sim t^{-\mu}$$

A broad spread of trap depths gives aging: escape gets harder with time.

Measured

Escape slope -0.77 in #51; negative in 6/8 periods. One directed message: odds $\times 1.5$ – 2.9 . More messages add little.

Implication

Kick a stuck agent early (hazard $\sim t^{-0.8}$). Send one message. For failure loops, use a reset.

Value per bit: only the context window has $\kappa_c > 0$

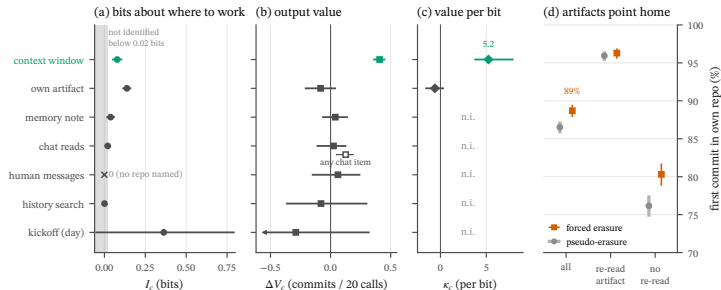
H87

credence 0.63

EU 1.57

models: 04 semantic info

tool 12 info dynamics



Model

$$\kappa_c = \frac{\Delta V_c}{I_c}, \quad \Delta V_c = \bar{V}(1 - e^{-b_2})$$

Each channel has a natural scramble;
value per bit κ_c ranks the channels.

Measured

Context: $\kappa_c = 5.2$ [3.8, 7.9]; a wipe costs 41% of output. Own artifact: most bits (0.14), but $\kappa_A = -0.6$ [-1.6, 0.3]. Memory, chat, search, human: value consistent with 0.

Implication

Protect the context window; a longer call cap pays. Notes and search do not buy it back. Test a status message after a wipe.

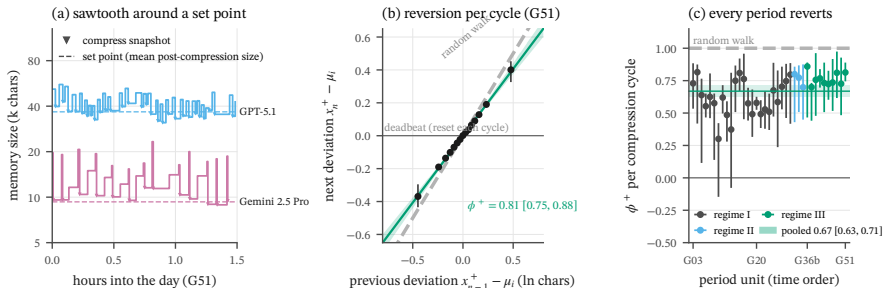
Memory size is an overdamped first-order homeostat

H71

credence 0.74

EU 1.11

models: 04 semantic info + 05 replicator dissipation



Model

$$x_n - \mu_i = \phi^+ (x_{n-1} - \mu_i) + \varepsilon_n$$

Each consolidation pulls log memory size back toward the agent's own set point μ_i .

Measured

$0 < \phi^+ < 1$ in 37/37 units; pooled 0.67 [0.63, 0.71]. No overshoot after forced wipes. Not first order: AR(2) +0.19, the target drifts slowly. H09's overshoot was a sampling artifact.

Implication

Set memory alarms per agent, not as one global cap. Use compression rows only. A prompt change moves the target, not the gain.

Identity: style and family fields

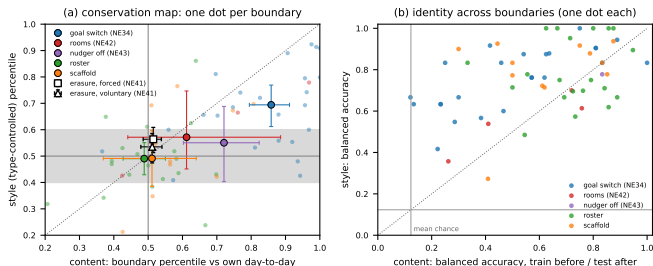
H46 Style is a context-dressed charge: not conserved, still identifying (EU 2.44)

H13 Family fields are a static style offset; no family-block coupling (EU 2.17)

Style is a context-dressed charge: not conserved, still identifying

H46 credence 0.75 EU 2.44

models: 11 vector spins + 04 semantic info



Model

$$T = \langle \text{rank} |\Delta s_{\text{bdry}}| \rangle, \quad T_{\text{conserved}} = \frac{1}{2}$$

A boundary (goal switch, wipe) is a collision. Does style survive it?

Measured

Style moves at goal switches and wipes. But it identifies agents at 0.76 vs 0.51 for content. It drifts as context fills.

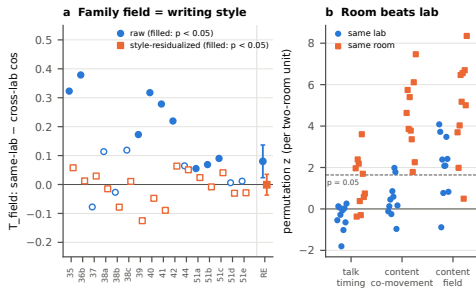
Implication

Use the 20 style rates to attribute messages or flag a model swap. Compare at the same context position.

Family fields are a static style offset; no family-block coupling

H13 credence 0.87 EU 2.17

models: 11 vector spins + 02 kinetic Ising + 01 inverse Ising + 10 Potts



Model

$$x_{i,d} = h_d + f_{\text{lab}(i)} + b_i + \xi_{i,d}$$

Is the family field f a position, or a style offset?

Measured

Family field in 9/15 units; after style removal, 0/9. No family coupling.
Room term $17\times$ the lab term.

Implication

Remove style before any embedding consensus metric. Do not infer vendor “ideology” from content.

Collective order: herding, allocation, culture

- H11** Potts coupling on projects is ferromagnetic in every goal mode (EU 2.06)
- H81** Regime-I content has a collective slow mode ($\tau \approx 25$ d) (EU 1.07)
- H35** The nudger is an inefficient Maxwell demon: bits buy activity, not work (EU 1.07)
- H89** Price decomposition: culture changes by transmission; selection ≈ 0 (EU 0.98)
- H94** Work allocation is a max-entropy state; ownership is the Lagrange multiplier (EU 0.92)

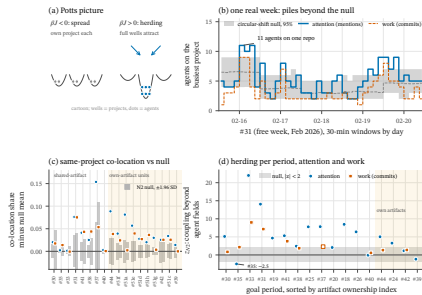
Potts coupling on projects is ferromagnetic in every goal mode

H11

credence 0.75

EU 2.06

models: **10 Potts** + 13 conventions + 15 stoch. thermo



Regime-I content has a collective slow mode ($\tau \approx 25$ d)

H81

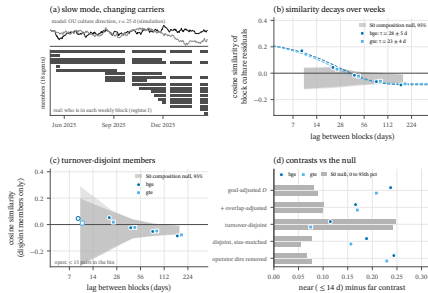
credence 0.61

EU 1.07

models: 11 vector spins

tool 14 scaling

+ 04 semantic info



Model

$$\langle u_b \cdot u_{b'} \rangle \propto e^{-|t_b - t_{b'}|/\tau_u}$$

Village content has a collective slow mode u_b beyond its members and goal fields.

Measured

Regime I: contrast 0.24 / 0.21 vs null 0.08 / 0.09; $\tau_u = 23\text{--}28$ d; it survives turnover. The artifact record does not carry it (power 1.0). Calendar-time drift is not excluded.

Implication

Compare a village with its own last 2–3 weeks. Curating artifacts or operator wording will not steer this mode.

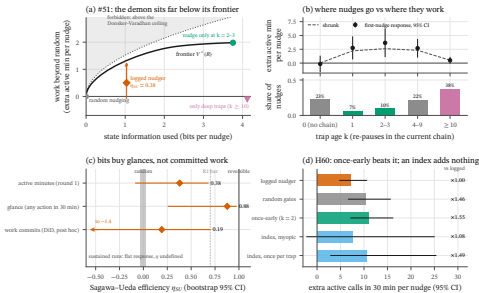
The nudger is an inefficient Maxwell demon: bits buy activity, not work

H35

credence 0.61

EU 1.07

models: 04 semantic info + 02 kinetic Ising



Model

$$\eta_{SU} = \frac{\Delta V}{V^*(I)}, \quad \Delta V \leq s\sqrt{2rl}$$

The nudger is a Maxwell demon: it spends bits of agent state to buy activity.

Measured

#51: 1.4 bits per nudge buy +1.45 active min, at 38% of the frontier. Commits +0.22/h [-0.29, 0.82], not significant. Switching the nudger on or off does not move swarm work.

Implication

Nudge each pause chain once, early ($k = 2-3$): $\times 2.3$ active minutes. Stop re-nudging deep traps. Do not expect more output.

Price decomposition: culture changes by transmission; selection ≈ 0

H89

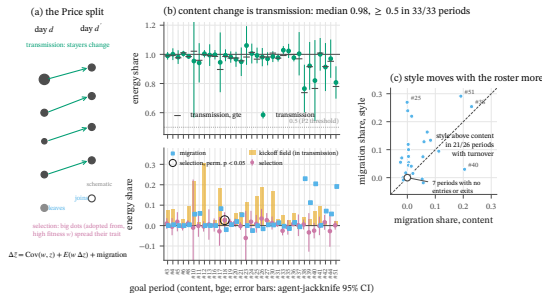
credence 0.78

EU 0.98

models: 13 conventions

+ 05 replicator dissipation

null 06 neutral drift



Model

$$\Delta \bar{z} = \text{Cov}_S(w, z) + E_S(w \Delta z) + \text{Mig}$$

Price equation: change of the mean trait = selection + transmission + migration.

Measured

Transmission carries 0.98 of the change (median, 33/33 periods). Selection 0.01: the design cannot detect it. Style migration exceeds content in 21/26. No attractor ($C > 0$ in 0/29).

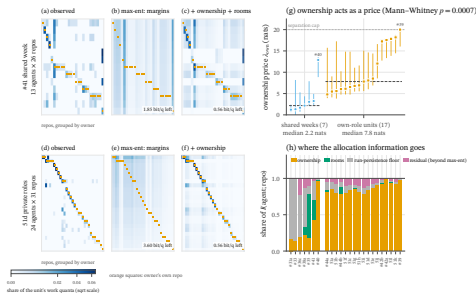
Implication

Read a content shift as a day field on all agents. A style shift on a join or exit day is mostly composition.

Work allocation is a max-entropy state; ownership is the Lagrange multiplier

H94 credence 0.61 EU 0.92

models: 10 Potts + 05 replicator dissipation vs 06 neutral drift



Model

$$P_{ij} \propto a_i b_j e^{\lambda_{\text{own}} \text{own}_{ij} + \lambda_{\text{room}} \text{room}_{ij}}$$

Work is a max-ent agent-repo table;
each constraint acts as a price λ .

Measured

Residual ≤ 0.13 of $I(\text{agent}; \text{repo})$ in
23/24 units. Ownership price 2.2 nats
in shared weeks, 7.8 in own-role units
($p = 0.0007$). No specialization beyond
ownership.

Implication

Track λ_{own} and κ each week. To
spread work, assign owners. To pool it,
name one shared repo in the kickoff.